



OPEN An LLM-based hybrid approach for enhanced automated essay scoring

John Atkinson[✉] & Diego Palma

Automated Essay Scoring systems have traditionally relied on shallow lexical data, such as word frequency and sentence length, to assess essays. However, these approaches neglect crucial factors like text structure and semantics, resulting in limited evaluations of coherence and quality. To address these limitations, we propose a hybrid approach to AES that combines multiple features from different linguistic levels. By leveraging the complementary nature of these features, our model captures the intricate relationships underlying coherent texts. Through extensive experimentation using standard essay datasets, we demonstrate that our large language model based hybrid model surpasses state-of-the-art methods based on shallow features and pure neural networks. This research represents a significant advancement towards the development of an accurate and effective tool for assessing student writing.

Keywords Automated essay scoring, Large language models, GPT, Natural-language processing, Neural context embeddings, Transformer

With the recent advances in computer technology, the possibility of grading essays with the assistance of computers has become a reality. Automated Essay Scoring (AES) has made it possible for teachers to assign scores to essays through computer analysis^{1,2}. AES programs operate by extracting a variety of features, such as word count, vocabulary choice, error density, sentence length variance, and paragraph structure from high-scoring essays to establish a model of essay quality³.

Essay assessment is a rigorous and time-consuming task for human graders. As such, text/essay assessment computational approaches have been developed in recent years to automate the process. However, many of these methods only use statistical features extracted from a text, such as word frequency, length, and sentence frequency. As a result, these methods can overlook the text's syntactical structure, leading to nonsensical essays that receive high ratings. Conversely, text coherence is often not accounted for, resulting in poorly structured or semantically phrased texts receiving good scores.

While all these approaches deal with coherence evaluation, they use shallow lexical features and/or lexical semantics with no syntax structure to assess a text's coherence. This, in turn, largely restricts the models' power to address a text's semantic and discourse properties that make it understandable in terms of syntactic, semantic and discourse connections, as a whole. Furthermore, many linguistic features complement each other, and when combined with context embeddings⁴, hidden relationships that underlie coherent texts can be taken into account. This hybrid model extends our past research on shallow linguistic features and discourse patterns by incorporating neural network context embeddings, resulting in an effective assessment of essay quality^{5,6}.

Unlike previous work, combining multiple features into different linguistic levels might be effective to grade an essay and be comparable to human performance. The rationale for this is that many linguistic features are complementary with each other and combining them with context embeddings can give account for hidden relationships that underlie coherent texts. Hence in this work, a hybrid model that combines shallow linguistic features, discourse patterns and neural context embeddings is proposed to effectively assess the quality of essays⁵.

Gold standards for a discourse are provided from essay datasets previously graded by human experts. A simple supervised classifier is then used to predict an essay score^{7,8}, ensuring that the model accurately assesses the quality of the essay. Our AES approach offers a more efficient and reliable way of grading essays, ultimately leading to improved student learning outcomes.

Accordingly, our main claim is that an AES method that considers shallow features discourse patterns and neural context embeddings can be more effective (i.e., well correlated with humans) than state-of-the-art approaches using shallow features or transformer-based neural learning.

Related work

Current AES approaches vary widely, from using traditional linguistic features and discourse analysis methods to purely deep neural network models that learn efficient low-dimensional context representations, known as

AI Empowered, Santiago, Chile. ✉email: john.atkinson@uai.cl

embeddings. However, using shallow linguistic features to evaluate essays has limitations, as it does not consider the overall cohesion and coherence of the essay. This can lead to highly rated but meaningless essays⁹.

Shallow features based methods

AES methods have usually been approached by extracting shallow linguistic features from an essay as indicators of its quality. It often includes the frequency of words and/or sentences, frequency of grammar errors, lexical categories, and readability indices^{10,11}, or even more complex indicators such as a text's lexical diversity regarded to as the used vocabulary. By using these features, an AES task can simply be seen as a regression problem task in these features are weighted so as to predict an essay's *score*, where the features' weights are estimated by collecting samples from human-assessed essays⁹.

One earlier popular system, *E-Rater*¹², can provide a holistic assessment for an essay as well as making real-time diagnostic feedback on grammar, word usage, style, and organization. Using this approach, human-model correlations greater than 0.8 were achieved. However, the method did not receive a proper acceptance in the educational community as using such simple and indirect measures, the approach would have a negative impact on the students, since they would concentrate more on satisfying the metrics rather than on producing a good essay^{13–18}.

Readability-based approaches

Some AES methods approach an essay's quality as a measure of its readability in which text difficulty is assessed by identifying grammatical and semantic features in the *Inter-agency Language Roundtable (ILR)* (<https://www.govtillr.org/>) scale. This language proficiency metrics can be used to assess the complexity of the language from no proficiency skills to native proficiency. Since the domain is a continuous spectrum, assessing a text's difficulty is seen as a regression problem that estimates the level of difficulty on the ILR scale (i.e., 0–5). In this approach, the *Morfessor Categories-Map* algorithm is applied for essay segmentation and uses relative entropy and statistical language models to measure its semantic complexity. Experiments showed that when compared to a traditional machine learning classifier such as Support Vector Machines⁸ the prediction errors differ by less than 0.07. The analysis considered three types of word usage features (i.e., word usage vectors), length measures (i.e., average length of words or sentences per document, relative entropy, average length of morphemes) and linguistic models (i.e., statistical language models according to the ILR level).

Coherence-based methods

In order to convey a communicative goal for a specific reader, a text must contain consistent and logical relationships between its utterances. Thus, a high-quality essay must be coherent, that is, it must have logical connections between units of information, both local (i.e., between adjacent statements) and global (i.e., all connections make sense in the full text—a discourse). In addition, a good essay should have high cohesion, that is, each sentence must be logically interpreted in relation to the others by using discourse markers, or semantic relationships connecting different sentences and paragraphs. Some coherence-based approaches use discourse patterns that can be extracted from semantic feature sequences contained in essays⁹. The baseline for a coherent discourse is usually provided by texts that have previously been assessed by human experts, which are passed on to an automatic classifier that can predict an essay's grade¹⁹.

Unlike previous approaches, this method combines latent space representation semantic coherence metrics and probabilistic entity grids-based discourse patterns in order to predict an essay's score^{16,19}. To this end, the method extracts three kinds of features:

1. *Shallow Features*: they are obtained by estimating metrics such as word frequencies, syllables, etc.
2. *Semantic Features*: they are extracted from GPT-3 embeddings for words and sentences.
3. *Discourse Patterns*: they are estimated by computing the transition probabilities from entity grids, which are generated from previously assessed essays. Patterns are then represented as vectors of entity transition probabilities.

A prediction model is then trained using those features from a training essay dataset to estimate the essay grade.

Discourse patterns based methods

This approach uses discourse pattern features to estimate an essay's coherence by combining two elements:

1. *Entity Grids (EG)*: Each essay is represented a two-dimensional matrix that captures the probabilistic distribution of discourse entities in an essay's sentences (aka. entity grids), in the form of transition patterns. This assumes that the distribution of the entities in coherent essays show some regularities in the corresponding EG so that the underlying essays will receive a better evaluation than those that do not show these regularities. Sample essays are divided into k sets of distinct evaluations. Cosine similarity is then calculated between a new essay and the test essay dataset, based on their probability transition vectors. This is the probability that, being in a state E_i at time n , it passes to state E_i at time $n + 1$.
2. *Semantic Metrics*: These are divided into two groups:
 - (a) *Lexicon*: it compares the lexical content of a new essay with that of human-assessed essay. Each essay is represented as a vector containing the frequency of the most relevant words.
 - (b) *Semantics*: it measures an essay's overall coherence as the semantic similarity between adjacent sentences using the semantic embeddings extracted from GPT-3.

The experiments used standard metrics such as weighted quadratic Kappa and Spearman's correlation²⁰ to determine the model-human rater correlation. These were compared against a reference method that uses shallow features. The results show a good correlation above 0.7 in most tests, outperforming shallow features-based approaches²¹. However, it's worth noting that traditional AES models are effective when evaluating essays on a specific topic, but when it comes to free text, their performance is reduced. One way to address this issue is by learning low-dimensional vector representations, such as word embeddings, that can be used to measure an essay's local coherence without the need for reference essays.

Neural context embedding based and LLM approaches

Traditional AES models are effective when evaluating essays on a specific topic. However, when a free text is approached, their performance is substantially reduced. One way to address this problem is by learning low-dimensional vector representations, aka. word embeddings that can later be used to compare adjacent units in an essay as a measure of its local coherence, without the need for reference essays.

Simple word embeddings can be learned from methods such as Word2vec, GloVe which use simple neural network architectures^{21–24}. One of the major problems with simple word embeddings methods is that they cannot capture complex contextual relationships in texts. In order to deal with these issues, deep neural network methods based on transformer architectures such as BERT, GPT, Llama have been introduced in order to train *large language models* (LLM)^{25–27}. Unlike previous word embeddings methods, LLMs give account for some semantic connections underlying the word/sentences by learning context embeddings using bi-directional relationships and focusing in the most prominent part of the sentences via attention mechanisms^{28,29}. Unlike previous approaches, fine-tune these LLMs on essay-specific datasets may be even more effective to the task of AES^{30–33}.

A significant issue for AES may be due to that LLM models even those handling large context windows cannot account for deeper semantic relationships and discourse patterns that underlie coherent essays. Furthermore, classical approaches are unable to generalize to new new topics in essays, especially in free texts. While transformers-based LLM are sensitive to the data they were trained on, they are well-suited for domain transfer tasks, mostly unseen in some NLP problems⁷.

Some of the features that have been used in recent approaches include lexical items, word embeddings^{4,22,34,35}, word categories⁶, readability difficulty of an essay, syntactical and semantic items, discourse features. The latter ones can encode an essay's discourse structure which may be produced by the following methods:

1. *Entity Grids (EG)*: these are probabilistic representations of discourse that are used to capture the local coherence of a text so that local coherence can be derived from estimating utterances' centers as they move along with a discourse.
2. *Discourse Analysis Trees*: these are based on rhetorical structures that encode a text's hierarchical discursive relations and are used to capture an essay's local and global coherence.
3. *Lexical Chains*: these are sequences of related words in a text, which are used as indicators of its cohesion. Intuitively, an essay containing many lexical chains tends to be more cohesive.

A hybrid model for AES combining LLM-based neural embeddings and shallow linguistic data

In the realm of AES, this research presents a novel hybrid model that propels the scoring process to new heights. By combining the strengths of shallow feature approaches and the improved accuracy of neural context embeddings derived from GPT-based models, our model offers an innovative solution for AES. This approach builds upon our prior work on extracting shallow linguistic features for AES¹⁹, extending it by incorporating state-of-the-art contextual embeddings from generative transformer-based architectures like GPT (Generative Pretrained Transformer). Through the integration of rich contextual information into a straightforward classifier, our model achieves unparalleled accuracy in predicting essay grades. To evaluate the efficacy of our model, a comparative analysis was conducted against a dataset of essays that had been previously assessed by human experts. The results yielded promising findings, establishing a high correlation between our model's predictions and human assessments. This work details the architecture and components of our model, as illustrated in Fig. 1.

Shallow features

The approach uses *shallow features* (SF) that allow evaluating the following aspects of an essay:

- *Readability*: it aims at estimating how easy is to read a text by using four metrics:
 1. *Gunning-Fog index*: it estimates the number of years of formal training a person would need to understand a text on a first reading, and is calculated as $Average\ length\ of\ a\ sentence + HW$ where HW is the number of words containing more than two syllables.
 2. *Flesh Index (FRE)*: it evaluates the readability of a text and is calculated as $206.835 - (1.015ASL) - (8.46ASW)$, where ASL is the Average Sentence Length and ASW is the Average number of Syllables per Word.
 3. *ARI Index*: it estimates the degree of text comprehension by measuring the number of linguistic features used and then estimating the training required to understand a text: $0.5(WPS) + 4.71(CPW) - 21.43$, where WPS is the average number of Words Per Sentence and CPW is the number of Characters Per word.

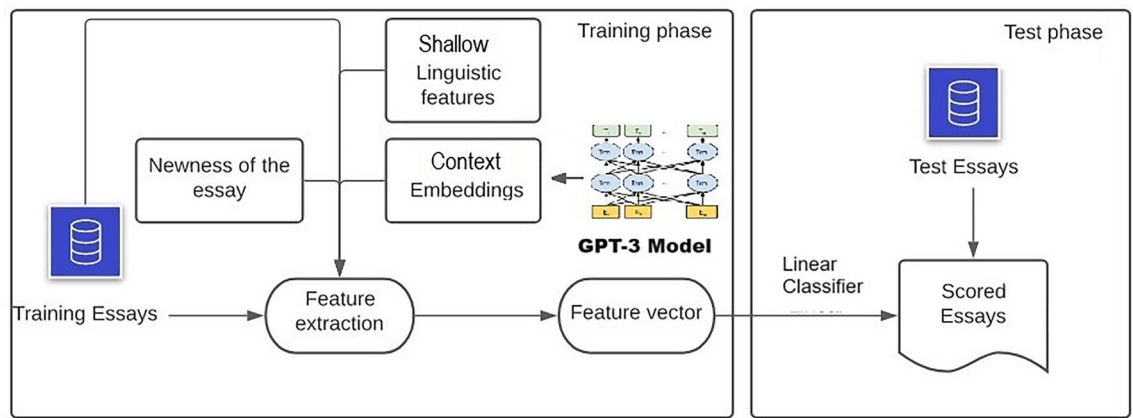


Fig. 1. Our GPT-based hybrid model for AES.

- **Lexical Diversity:** it estimates the diversity of the vocabulary used by a writer in an essay and can be obtained using the following metrics. Diversity is accomplished by using the *Guiraud Index (GI)* which penalizes a text depending on its length (e.g., the longer, the worse) and is calculated as $\frac{v}{\sqrt{N}}$, where v is the number of distinct words and N the number of words.
- **Grammar:** it aims at measuring the efficient use of linguistic resources such as lexical categories, n-gram sequences, frequency of words that belong to some lexical category (i.e., verbs, nouns).
- **Complexity:** Our novel approach to AES sets itself apart from previous methods by not only grading an essay's coherence but also assessing its complexity. Our model estimates an essay's degree of information gain by analyzing adjacent units such as sentences or noun phrases (NPs) and determining the amount of *newness* or *givenness* they convey. This provides a more holistic evaluation of an essay's overall quality, taking into account its level of complexity. By incorporating this additional feature, our model offers a more comprehensive and nuanced analysis of essays that is unmatched by other AES methods.
- Accordingly, the components of an essay can be grouped into three classes: given, partially given (i.e., based on various types of inferential availability), and not given (i.e., newness). When assessing a text, it becomes more natural to include new information as that is not given. Hence the complexity should first measure how much *givenness* is contained in an essay and then estimate the remaining information as *newness*. In addition, two additional metrics are considered to assess a text complexity:
 - **Givenness:** in order to estimate this metric, sentences of an essay are converted into a vector space representation. An hyperplane of all the sentence vectors but the target sentence (G) are then built. The current sentence vector is projected onto that hyperplane. This projection is seen as a vector component shared with a previous essay whereas the vector component that is orthogonal to the hyperplane is seen as a new sentence (N). *Givenness* is then calculated as $\frac{N}{(N+G)}$
 - **Pronoun Density:** An essay with a higher density of pronouns will be more difficult to read than a text with a lower density of pronouns as more comprehension inferences must be made. It can be estimated by calculating the proportion between the pronouns (PN) and the total number of words (N_{words}) an essay: $\frac{PN}{(N_{words})}$.
 - **Pronoun-Noun relationship:** Since pronouns are co-references to a previously introduced proper name or noun, this relationship becomes a measure of complexity of reading the text. Thus, it can be estimated as the ratio between the number of pronouns (PN) and the total number of nouns (TN), that is, $\frac{PN}{TN}$.

Neural context embeddings from generative models

One of the problems with traditional embedding generation models is that vectors representing meaning are static and unique regardless of the context where words are used. Therefore, different meanings of a word share the same representation, which bring serious problems with dealing with polysemy and synonymy. One way to address this issue is by learning low-dimensional vectors, aka. *embeddings* using transformer-based generative models such as GPT, which allow learning context-sensitive word representations. Unlike previous approaches to AES¹⁹, our model uses neural context embeddings as a major feature to measure local coherence based on the similarity between the representation of adjacent sentences.

Unlike BERT and other recent LLMs, generative models like GPT offers several advantages for our target problem:

- **Unidirectional Context:** these generate text in a unidirectional manner. This allows them to generate coherent and contextually appropriate responses to prompts, making them suitable for tasks where understanding the context of the entire input is crucial, such as essay writing.

- *Generative Capability*: these are generative, meaning they can produce human-like text based on a given prompt. This capability is particularly useful for tasks that require producing language output, such as text completion and essay production.
- *Longer Contextual Understanding*: these have a broader contextual understanding over longer passages compared to BERT. This enables them to capture more nuanced relationships and dependencies within the text, leading to better performance on tasks that require understanding complex language structures.
- *Scalability*: these can be scaled up to include a larger number of parameters, which has been shown to improve performance on various NLP tasks.

In our hybrid model for AES, these neural context embeddings are learnt from a decoder-only transformer provided by GPT as described in Fig. 2. GPT operates by first pretraining on vast amounts of text data using unsupervised learning, where it learns to predict the next word in a sequence given the preceding words. This pretraining process enables GPT to capture general language patterns and context from diverse text sources. GPT employs a transformer architecture with self-attention mechanisms, allowing it to effectively weigh the importance of different words in the input sequence and capture long-range dependencies. Stacked transformer layers refine the learned representations, enabling the model to capture increasingly complex linguistic patterns. During fine-tuning on specific tasks, GPT adapts its pretrained knowledge to the requirements of the target domain, optimizing its performance for a target task.

In order to generate neural embeddings, GPT first tokenizes input text into subword units or tokens and then representing each token as an embedding vector (Fig. 2). These embeddings capture the semantic meaning of the tokens within the context of the entire input sequence. GPT incorporates positional encoding to provide information about the position of tokens within the sequence, allowing the model to distinguish between tokens at different positions and capture their relative positions accurately. The embeddings are generated through a series of transformer layers, where each layer refines the representations learned from the previous layers. Through self-attention mechanisms, GPT effectively weighs the importance of different tokens in the input sequence and captures long-range dependencies, leading to rich and contextualized embeddings that encode the semantic and positional information of the input text effectively.

Experiment and results

In order to assess the efficacy of our hybrid model for AES, we conducted a rigorous evaluation by implementing diverse shallow feature (SF) extraction techniques in conjunction with a pretrained GPT-3 model for contextual embeddings learning. To validate our model's effectiveness, we leveraged a range of standard metrics and benchmarked it against datasets of human-annotated essays. We set our model apart by comparing it with established state-of-the-art AES techniques such as discourse patterns (DP), *Semantic Coherence* (SC), and GPT-only approaches. This comparison allowed us to demonstrate the superiority of our hybrid model over existing methods in terms of precision, recall, and other critical evaluation metrics.

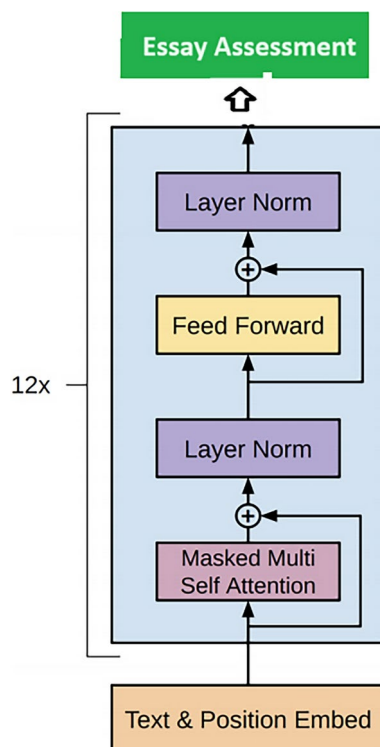


Fig. 2. A Decoder-only architecture in GPT for essay scoring.

ID	Type	Grade level	Size of training dataset (essays)	Size of test dataset (essays)	Average length (words)	Score range
1	PER, NAR, EXP	8	1785	592	350	2–12
2	PER, NAR, EXP	10	1800	600	350	1–6, 1–4
3	SDR	10	1726	575	150	0–3
4	SDR	10	1772	589	150	0–3
5	SDR	8	1805	601	150	0–4
6	SDR	10	1800	600	150	0–4
7	PER, NAR, EXP	7	1730	576	250	0–30
8	PER, NAR, EXP	10	918	305	650	0–60

Table 1. Essays dataset description.

	1	2a	2b	3	4	5	6	7	8	Avg.
Our model	0.72	0.79	0.81	0.75	0.76	0.77	0.76	0.29	0.38	0.67
DP	0.64	0.77	0.76	0.71	0.74	0.74	0.73	0.26	0.35	0.63
SC	0.44	0.66	0.66	0.60	0.55	0.59	0.56	0.17	0.17	0.48
GPT-3	0.41	0.62	0.74	0.68	0.67	0.70	0.68	0.22	0.25	0.55

Table 2. EA for different AES approaches.

Essay collection

Several essay corpora from AES competitions available at Kaggle (<https://www.kaggle.com/c/asap-aes>) were collected. These are organized into four essay categories: Persuasive (PER), Narrative (NAR), Expository (EXP), and Source-Dependent Response (SDR). Essays were written by high school students (8–10th degree) and manually scored by humans, except dataset no. 2, which was rated using different criteria (Table 1).

It is important to highlight that essays dataset 3, 4, 5, and 6 are not free-form texts but the result of a script answering a question based on a given source. Hence, this type of essay has a higher semantic closeness to a question/query compared to free-form texts on a topic.

Linguistic feature extraction

A set of shallow features, metrics of essay complexity, and GPT-3 embeddings were extracted to create a feature vector that can be used to predict an essay's final grade. This included *Shallow Features* (i.e., Readability and lexical diversity indices, and grammatical features), *Essay's complexity* which considered *Pronoun Density* and *Pronoun-Noun Relationship*, and *Context Embeddings* from GPT-3.

Predicting an essay's score

To generate the prediction model, a simple model based on *Random Forests* was trained and used. To this end, the dataset was divided into a training set and a test set using cross-validation methods such as K-fold ($k = 10$), using the *Random Forest Regression* and *Random Forest Classifier* implementations in Python.

Assessing the hybrid AES model

The performance of our approach was measured by using four standard metrics:

1. *Spearman Correlation (r)*: it measures the correlation between the model and a human evaluator.
2. *Exact Agreement (EA)*: it evaluates the proportion of essays that were assessed with the same grade as a human evaluator.
3. *Adjacent Agreement (AA)*: it measures the proportion of essays that were assessed with a maximum of one score of difference with that of a human evaluator.
4. *Quadratic Weighted Kappa (QWK)*: it evaluates the level of agreement between two raters who classify items into an ordinal scale. This is specially important for our dataset as the majority of score range is small so AA is not enough for the assessment.

Firstly, experiments were conducted to compare our model against two main approaches: (1) Shallow Features (SF) + Discourse Patterns (DP) and (2) GPT + SF as shown in Tables 2, 3, and 4 respectively for all the metrics.

The results show that our model outperforms the approaches based on PD+SF and GPT+SF on the QWK metric, with a significant difference in essays 1, 2a, 2b and 8. In general, the average QWK of our approach is higher than those of the other approaches, indicating a higher correlation with human evaluation.

Our hybrid model for AES has shown significant improvements in the assessment of Persuasive, Narrative and Expository essays when compared to state-of-the-art approaches such as *discourse patterns* (DP), local semantic coherence (|) and pure GPT-based methods. This can be partially attributed to GPT's focus on the contextual meaning of words, enabling it to better evaluate essays with greater writing flexibility. However, the limitations of GPT's inability to generate embeddings for words not appearing in the training dataset emphasizes

	1	2a	2b	3	4	5	6	7	8	Avg.
Our model	0.99	1.0	1.0	0.99	0.99	0.99	0.99	0.82	0.86	0.96
DP	0.99	1.0	1.0	0.98	0.99	0.99	1.0	0.81	0.86	0.96
SC	0.94	1.0	1.0	0.98	0.99	0.99	0.99	0.71	0.72	0.92
GPT-3	0.68	1.0	1.0	0.98	0.99	0.99	0.99	0.78	0.86	0.92

Table 3. AA for different AES approaches.

	1	2a	2b	3	4	5	6	7	8	Avg.
Our model	0.91	0.79	0.77	0.71	0.84	0.84	0.84	0.87	0.92	0.83
DP	0.88	0.77	0.71	0.69	0.84	0.84	0.84	0.85	0.90	0.81
SC	0.68	0.55	0.60	0.45	0.52	0.65	0.48	0.66	0.66	0.58
GPT-3	0.75	0.62	0.69	0.66	0.74	0.79	0.78	0.82	0.56	0.71

Table 4. QWK for different AES approaches.

QWK	EA	AA
0.83	0.84	1.0

Table 5. Results of fine-tuned GPT-3 on the essay dataset.

	1	2a	2b	3	4	5	6	7	8	Avg.
SAGE	0.93	0.79	0.67	0.83	0.81	0.87	0.78	0.88	0.81	0.82
PEG	0.82	0.72	0.70	0.75	0.82	0.83	0.81	0.84	0.73	0.78
Our model	0.91	0.79	0.77	0.71	0.84	0.84	0.84	0.87	0.92	0.83
e-rater	0.82	0.74	0.69	0.72	0.80	0.81	0.75	0.81	0.70	0.76
IntelliMetric	0.78	0.70	0.68	0.73	0.79	0.83	0.76	0.81	0.68	0.75
CRASE	0.76	0.72	0.69	0.73	0.76	0.78	0.78	0.80	0.68	0.74
LightSIDE	0.79	0.70	0.63	0.74	0.81	0.81	0.75	0.77	0.65	0.74
AutoScore	0.78	0.68	0.66	0.72	0.75	0.82	0.76	0.67	0.69	0.73
IEA	0.79	0.70	0.65	0.68	0.77	0.81	0.80	0.78	0.72	0.73
Lexile	0.66	0.62	0.55	0.65	0.67	0.64	0.65	0.58	0.63	0.63

Table 6. QWK for different state-of-the-Art AES systems. Significant values are in bold.

the importance of incorporating shallow feature (SF) extraction techniques to supplement the contextual information provided by GPT.

Additionally, our hybrid model exhibits a slight drop in performance for essays that require a specific answer to a given question. Here, SF-based approaches prove to be more effective than our approach in terms of the QWK metric. This is because there is only one correct answer for such essays, leaving little room for the student's creativity and interpretation.

However, for the *Exact Agreement* (EA) and *Adjacent Agreement* (AA) metrics, our approach demonstrates significant improvements over other AES models. It is noteworthy that for some essay sets, the grade ranges between 1 and 60, making it difficult for any model to obtain an exact score. Despite this, our hybrid model outperforms GPT + SF and DP + SF approaches in these challenging sets. In order to investigate the extent to which a GPT-3 fine-tuned model may impact the performance, the metrics were also recomputed for the whole essay dataset using normalized scores as seen in Table 5.

Furthermore, our approach was compared against popular state-of-the-art AES systems, where it outperformed them on the QWK metric, as shown in Table 6. These results demonstrate the effectiveness and superiority of our hybrid model for AES compared to both academic and commercial models.

Our hybrid method for AES has demonstrated better performance over traditional statistical or SF approaches, as well as commercial systems such as Lexile, Brookete, LightSide, and Autoscore. These systems rely on shallow features and readability measures to predict an essay grade, which limits their ability to capture contextual information and to deal with polysemy and synonymy.

Although SAGE outperforms our approach in several sets of essays, our method significantly outperforms SAGE on datasets where no contextual knowledge can be inferred. SAGE relies on multiple centrality-based measures of textual coherence and ontologies to determine whether certain essay clauses are correct or incorrect.

However, our approach's use of contextual representations provides a deeper sense of language context and flow, allowing it to better assess essays where there is greater freedom of writing.

The proposed approach, which combines shallow features and neural contextual embeddings, outperforms several state-of-the-art approaches AES systems for different kinds of essays. Only two working systems, SAGE and PEG, outperform our approach due to their use of additional domain knowledge bases. These results demonstrate the effectiveness of our hybrid approach and its potential to contribute to the development of more accurate and reliable AES systems.

Conclusions

This work introduced a novel hybrid method for AES that combines the power of shallow linguistic features and generative neural context embeddings via LLM models such as GPT-3. The proposed AES method represents a significant advancement in essay grading, offering a comprehensive and accurate assessment of essay quality by considering various factors such as readability, lexical and grammatical diversity, and reading complexity. By incorporating word contexts, the approach effectively evaluates essays with greater writing freedom.

Through extensive experimentation, our results suggested that the proposed hybrid approach surpasses numerous state-of-the-art methods and even outperforms several AES systems in terms of human-model correlation. While some systems, like SAGE, leverage additional knowledge bases for further inferences that our approach currently lacks, our hybrid method remains competitive with notable AES systems including SAGE, PEG, and e-rater in terms of correlation and agreement metrics.

The findings suggest that the proposed hybrid approach exhibits high effectiveness in assessing diverse types of essays. Particularly, in datasets with wider grade value ranges, reaching up to 60, our approach achieved agreement rates exceeding 70%, significantly surpassing other systems. Overall, the proposed AES method offers an accurate and reliable assessment of essay quality, rendering it a valuable tool for educational institutions, writing centers, and individuals aiming to enhance their writing skills.

Future research endeavors could focus on exploring additional knowledge bases or integrating external resources to further enhance the capabilities of our hybrid approach. Moreover, investigations into addressing clauses' truth inference, similar to systems like SAGE, would contribute to expanding the scope of our method. With continuous advancements, the proposed hybrid AES method holds the potential to reshape the landscape of AES, enabling more effective and efficient evaluations in educational settings.

Data availability

The data that support the findings of this study are available from the corresponding author, J. Atkinson, upon reasonable request.

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Author contributions

All authors have contributed to this research: 1. made substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data; or the creation of new software used in the work; 2. drafted the work or revised it critically for important intellectual content; 3. approved the version to be published; and 4. agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Declarations

Competing interests

The authors report there are no competing interests to declare.

Additional information

Correspondence and requests for materials should be addressed to J.A.

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